

Project Overview

Welcome to the core documentation and feature breakdown for our 3D OpenGL FPS game project. Developed entirely using customized demo frameworks and low-level rendering functions, except some functions that we felt like we needed to bring to the table in order to make the game more playable and less frustrating to play, also implemented basic sound effects using pygame library. Below is the comprehensive architectural overview, control scheme, and complete feature list governing the game.

Controls & Input Scheme

- **Movement:** W, A, S, D (Standard First-Person WASD Navigation with dynamic view bobbing).
- **Look / Aim:** Mouse Look (Precise viewport orientation).
- **Shoot / Fire (MB1):** Hold-to-charge mechanics for magic spells/arcane staff/bow then release to fire, or standard firing for crossbow.
- **Deployable Ability (E):** Deploys active field turret onto the field that supports the players by constantly attacking the enemy, removes and picks it up if Pressed E again near the turret.
- **Reload Button (R):** If the crossbow is empty, then reload the crossbow.
- **Pause Button (P):** Pauses the game.
- **Reset Button (R):** Restarts the game whenever dead and its game over.
- **Switch Weapons (1, 2, 3, 4):** Cycles through the 4 available weapons.
- **Dash (Spacebar):** Dashes upfront if pressed the button, a 2 second cooldown starts.
- **Inventory & Upgrades:** Interactive Shop / Runes / Stat management (Using mouse and clicking to purchase upgrades from shop).
- **Entering Shop (B or F):** When close to a shop, pressing the button allows us to access the shop terminal, only available during wave breaks, not available while a wave is active (or enemies are alive).
- **Main Menu Navigation:** Can be done using mouse and clicking.

Comprehensive Feature List

Player Combat & Weaponry Systems

- **Arcane Staff:** Long-range magic weapon featuring a hold-to-charge mechanic holding fire button (MB1) that scales up projectile size and damage output, consuming mana in the process. Upon hit, the projectiles cause Area-of-effect damage up to a certain radius. Can be upgraded in the shop to increase weapon damage.

- **Hand Magic:** Close-to-mid range elemental spellcasting, utilizing a similar charge-up mechanic holding fire button (MB1) for high-impact blasts, complete with mana consumption rules, however it consumes less mana than arcane staff but also does less damage. Upon hit, the projectiles cause Area-of-effect damage up to a certain radius. Can be upgraded in the shop to increase weapon damage.
- **Crossbow:** Slow reload time paired with instant-fire capability, high single-target damage, pinpoint accuracy, and long-range effectiveness. Can be upgraded in the shop to increase weapon damage and reload speed.
- **Bow:** Fast reload mechanics with a slower firing cadence, medium damage output, medium accuracy. Can be upgraded in the shop to increase weapon damage.
- **Mana & Elixirs:** Energy resource management supporting magic weapons, coupled with a Mana, health elixirs, and temporary Invincibility power-ups.
- **Deployable Turret:** A turret set up by the player to target enemies. Its visuals change depending on the level of turret (5 distinct levels), can be upgraded from the shop, its fire rate and damage increase with level.

World Hazards, Economy & Interactive Zones

- **Shops:** Shops that are across the game map allowing players to purchase upgrades for both Deployable Turret (can also purchase additional turrets, max count is 3) and the full weaponry system, and restoration items like mana, health and invincibility elixir.
- **Spiked Floor Traps:** Hazard zones placed across the environment that trigger damage and mana drains upon contact.
- **Magic Beam Traps:** A rotating beam from a pillar which progressively damages players and reduces mana.
- **Glowing Runes:** Collectible health and mana runes scattered throughout the map to sustain the player during intense combat waves.

Enemy AI & Combat Encounters

- **Archer Enemies:** Ranged combatants that maintain distance and punish the player with projectile or proximity pressure.
- **Regular Enemies:** Standard frontline combat units swarming the player with direct melee pressure.
- **Heavy Enemies:** Slower, heavily armored brawlers possessing expanded hitboxes and high health pools.
- **Sage Enemies:** Specialized magic-casting units that buffs up the enemies and heals them.

Visual Effects & Animations

- **Dynamic Weapon Animations:** Responsive visual feedback including weapon lowering during reloads, weapon recoil kickback on the bow and crossbow, and forward-thrust animations when casting hand or staff magic.
- **Immersive View Bobbing:** Procedural camera movement synced to player walking states that locks instantly to zero when stationary.
- **Spell & AOE Visual Effects:** Custom particle rings and color-coded visual effects mapped directly to magic attacks, arrow hits, and elemental impacts.

Sound Effects

- **For most of the actions in the game, a sound will play, it includes:**
 1. **Buying items and Using Runes/Elixir**
 2. **Shooting weapons (each weapon has their individual sound effects, including pulling a bow, and releasing it which has separate sound effects)**
 3. **Hurt Noises**
 4. **Deployable turret sound (Sound plays whenever the turret is deployed, picked up or it fires).**
 5. **When there's no ammo, a sound plays, if we still try to fire it.**